Course Three

Go Beyond the Numbers: Translate Data into Insights

Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- Complete the questions in the Course 3 PACE strategy document
- Answer the questions in the Jupyter notebook project file
- Clean your data, perform exploratory data analysis (EDA)
- Create data visualizations
- Create an executive summary to share your results

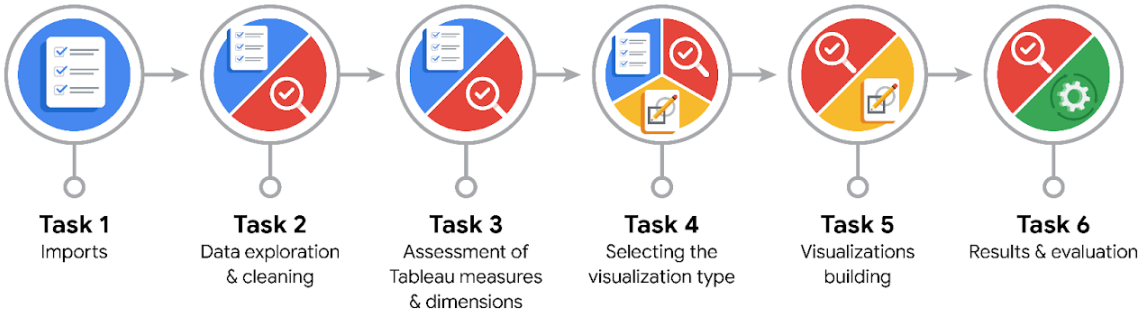
Relevant Interview Questions

Completing the end-of-course project will help you respond these types of questions that are often asked during the interview process:

- How would you explain the difference between qualitative and quantitative data sources?
- Describe the difference between structured and unstructured data.
- Why is it important to do exploratory data analysis?
- How would you perform EDA on a given dataset?
- How do you create or alter a visualization based on different audiences?
- How do you avoid bias and ensure accessibility in a data visualization?
- How does data visualization inform your EDA?

Reference Guide

This project has six tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- What are the data columns and variables and which ones are most relevant to your deliverable?

trip_distance and total_amount.

- What units are your variables in?

Datetime, integer, floats and categorical.

- What are your initial presumptions about the data that can inform your EDA, knowing you will need to confirm or deny with your future findings?

That some of the variables distributions will be right-skewed and contain extreme values/outliers and some monetary variables having negative amounts.

- Is there any missing or incomplete data?

No there is no missing or incomplete data within the dataset.

- Are all pieces of this dataset in the same format?

No, there is categorical and numerical as well as datetime objects in the data.

- Which EDA practices will be required to begin this project?

Discovering, Structuring, Cleaning, Validating and Presenting.

PACE: Analyze Stage

- What steps need to be taken to perform EDA in the most effective way to achieve the project goal?

Making sure to validate everything that is done within the EDA as to make sure the insights gained are reliable and accurate. Additionally, to visualize the distributions, relationships and outliers in the dataset.

- Do you need to add more data using the EDA practice of joining? What type of structuring needs to be done to this dataset, such as filtering, sorting, etc.?

No additional data needs to be added via the practice of joining as the dataset is complete. The structuring needs that are to be done is sorting and filtering with Boolean masking, as well as converting the datetime variables to datetime objects. Additionally, subsets of the data is to be made with groupby statements, Boolean masking and sorting for visualization and analysis.

- What initial assumptions do you have about the types of visualizations that might best be suited for the intended audience?

The initial assumptions that I have are that box plots, histograms, bar charts and scatter plots will be useful to showcase the distribution of the data, outliers and relationships between variables.

PACE: Construct Stage

- What data visualizations, machine learning algorithms, or other data outputs will need to be built in order to complete the project goals?

Variable distributions via histograms and box plots, the relationship between total_amount and trip_distance as a scatter plot as well as bar charts showing the relationship between different variables such as monthly total revenue.

- What processes need to be performed in order to build the necessary data visualizations?

The processes that need to be carried out for the data visualizations are making subsets of the data of groupby statements, Boolean masking, sorting and the conversion to datetime objects.

- Which variables are most applicable for the visualizations in this data project?

total_amount, tip_amount and trip_distance.

- Going back to the Plan stage, how do you plan to deal with the missing data (if any)?

If there is missing data, I would first investigate the cause of this missing data and either fix it via upsampling or downsampling, or leave it if it does not create a huge data imbalance.

PACE: Execute Stage

- What key insights emerged from your EDA and visualizations(s)?

I have learned that there are many outlier/extreme values in the dataset as well as right-skew distributions. Additionally, the tip amount across the different vendors had the same distribution across the vendor types relative to the vendor types proportion in the dataset, with this distribution being maintained even at larger tip amounts. The average tip amount by passenger count varies slightly by the passenger count with their being a slight dip with 4 passengers, but this is expected due to the low passenger count of 4 passengers in the dataset. The total rides per month was fairly

consistent with March and October having the most monthly rides and drops in July, August, September and February. I contrast, the total rides per day had Wednesday through to Saturday having the most amount of daily rides with Sunday and Monday having the least. Total revenue by day had Thursday with the highest amount of total revenue with Sunday and Monday having the least. Total revenue by month showed a similar pattern to the monthly total rides with drops occurring in July, August, September and February, with May having the highest amount of total revenue. Finally, the drop-off locations are evenly distributed over the area of New York with the Histogram showing that some of the locations receive a disproportionate amount of drop-offs compared to the other locations and this could be due to common trips such as to the airport or tourist destinations.

- What business and/or organizational recommendations do you propose based on the visualization(s) built?

To investigate the outliers and extreme values and decide on whether to reassign, leave or remove them and to conduct hypothesis testing to check for any statistically significant relationship between two variables.

- Given what you know about the data and the visualizations you were using, what other questions could you research for the team?

Why is May and Thursday the highest amount of total revenue? Why are there trip distances of 0 and what could they be? Why are there extreme values in trip distance such as over 33 miles? What do the location IDs refer to? Why is there a drop in total revenue between Thursday and Saturday when there is a small difference in their total rides?

- How might you share these visualizations with different audiences?

By using design thinking and making sure that the visualisations are accessible to all such as with vision impairments. Additionally, to make the visualisations less confusing and more self explainable for non-technical audiences.